

Accelerometer-Based Gesture-Controlled Robot

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Presented here is a gesture-controlled robot using AT89C52—one of the most basic microcontrollers. The robot can be moved in any direction just by making simple gestures, and the system's sensitivity to gestures can be easily adjusted as

per your liking. The LCD on the robot indicates the present status of the robot's movement.

Circuit and working

Fig. 2 shows the block diagram of the complete project. The movement of gesture transmitter section is sensed by the accelerometer and the voltage

levels corresponding to movement of the robot along X and Y axes are compared by a comparator, following which corresponding instructions are transmitted through the RF transmitter to control the robot. The robot receives the instructions. The microcontroller drives motors corresponding to the instructions received and displays the related message on the LCD.

Fig. 3 shows the circuit of the gesture transmitter. It is built around encoder HT12E (IC1), precision voltage comparator LM339 (IC2), regulator 7805 (IC3), RF transmitter module, accelerometer module and a few discrete components.

Pins 9 and 10 of comparator IC2 are connected to X-axis output pin of the accelerometer. Voltage levels corresponding to the motion along X-axis are continuously compared with a reference voltage set by presets VR3 and VR4 using op-amps A3 and A4. Similarly for Y-axis, voltage levels corresponding to the motion along Y-axis are continuously compared with a reference voltage set by presets VR1 and VR2 using op-amps A1 and A2. If the voltage level for any axis crosses the predefined voltage level, the corresponding op-amp's output goes low, generating a different four-bit code from the output of all four op-amps of IC2 for motion along each axis.

This four-bit code is fed to pins AD8 through AD11 of encoder IC1.

PARTS LIST

Semiconductors:

- IC1 - HT12E encoder
- IC2 - LM339 comparator
- IC3, IC4 - 7805, 5V regulator
- IC5 - HT12D decoder
- IC6 - AT89C52 microcontroller
- IC7 - L293D motor driver
- T1 - BC547 npn transistor
- LED1-LED6 - 5mm LED
- D1, D2 - 1N4007 rectifier diode
- RF TX1 - ASK transmitter module
- RF RX1 - ASK receiver module
- Accelerometer module

Resistors (all 1/4-watt):

- R1 - 1-mega-ohm
- R2-R5 - 470-ohm
- R6, R7 - 330-ohm
- R8 - 51-kilo-ohm
- R9 - 1-kilo-ohm
- R10 - 10-kilo-ohm
- VR1-VR5 - 10-kilo-ohm preset

Capacitors:

- C1, C2 - 0.1µF ceramic disk
- C3 - 1µF, 16V electrolytic
- C4, C5 - 33pF ceramic disk

Miscellaneous:

- DIP1, DIP2 - 16-pin DIP switch
- S1, S2 - On/off switch
- S3 - Push-to-on switch
- M1, M2 - 12V, 100-rpm geared DC motor
- BATT.1 - 9V battery (PP3)
- BATT.2 - 12V, 4.5Ah battery
- ANT.1-ANT.2 - 10cm hookup wire antenna
- LCD - 16x2 LCD module
- X₁ - 11.0592 MHz crystal oscillator
- Accelerometer module

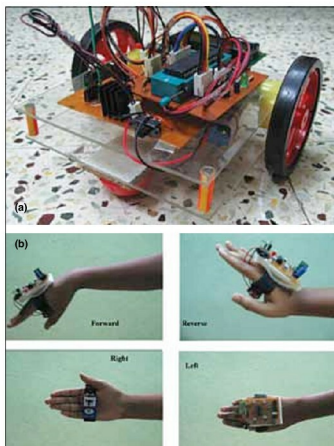


Fig. 1: (a) Author's prototype and (b) various hand gestures made to move the robot through the controller

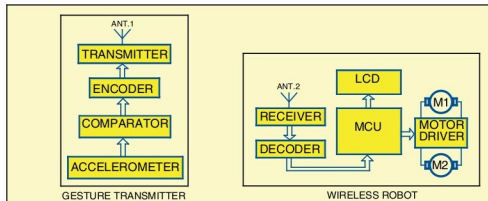


Fig. 2: Block diagram